Assignment 6B. Simpson's and Monte Carlo method Marks 20

Posted on 16.09.2026 @ 2:30 pm and due on 16.09.2026 @ 5:40 pm

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1. Write library function for Simpson's 1/3-rule and Monte Carlo integration by passing function. **[10]**

2. Evaluate the following integrals using Midpoint and Simpson's 1/3-rule accurate up to 6 places in decimal. Use the error bound formulae to determine N. **[3+3]**

$$\int_1^2 \frac{dx}{x} = 0.69314718, \qquad \int_0^{\pi/2} x \cos dx = \frac{\pi}{2} - 1$$

3. Use Monte Carlo integration scheme (using your own pRNG) to estimate the integral accurate up to 4 places in decimal **[4]**

$$\int_{-1}^{1} \sin^2 x \, dx$$